Project Domains (Choose One)

Learners may choose from the following or propose a custom domain (with approval):

- **Healthcare:** Predict patient readmission or disease likelihood
- **Finance:** Detect fraudulent transactions
- **eCommerce:** Recommend products to users
- **Education:** Predict student dropout risk
- **Cybersecurity:** Detect anomalies in network traffic
- **Clustering Option (Unsupervised):** Group customers, behaviours, or products based on similarity (K-Means, DBSCAN, Hierarchical)

Step-by-Step Instructions

Step 1: Problem Understanding & Framing

- Frame the business and data science problem clearly.
- Define whether it's a classification, regression, recommendation, anomaly detection, or clustering task.
- Specify success metrics (e.g., Accuracy, AUC, RMSE, Silhouette Score) and business KPIs (e.g., cost savings, uplift).
- **Capstone linkage:** Module 1 output maps to Capstone Steps 1–3.

Deliverable: Clear problem statement + task type + target metric.

Step 2: Data Collection & Understanding

- Use public datasets (Kaggle, UCI, APIs, etc.) or approved custom data.
- Summarise feature types, missing values, outliers, etc.
- Provide a **data dictionary** (variables, types, units, allowed values).

Deliverable: Dataset overview + data dictionary.

Step 3: Data Preprocessing, Applied EDA & Feature Engineering

- **Clean data:** Handle nulls, duplicates and outliers.
- **Engineer features:** Scaling, encoding, binning and domain-derived features.
- **Applied EDA**: Distributions, relationships, clustering tendency (if unsupervised).

- **Feature importance & explainability**: SHAP, LIME, or model-based importances.
- **Feature selection**: At least one approach (filter, wrapper, or embedded).
- **Dimensionality reduction**: PCA (and t-SNE/UMAP for visualisation if needed).

Deliverable: "EDA + Feature Engineering Report" with reproducible code & justifications.

Step 4: Model Implementation

- Experiment with appropriate models:
    - **Supervised**: Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, etc.
    - **Unsupervised**: K-Means, DBSCAN, Hierarchical (Elbow, Silhouette).
    - **Recommendation**: Collaborative or content-based.
    - **Deep Learning**: RNNs, CNNs, LSTMs, Transformers (if appropriate).
- Evaluation: compare with relevant metrics.
- **Reproducibility**: save configs and artefacts (models/).

Deliverables: Trained models, metrics and comparison between models.

Step 5: Critical Thinking → Ethical AI & Bias Auditing

- Explain model decisions (SHAP, LIME, PDP, ICE).
- Address limitations (imbalance, leakage, overfitting).
- **Bias detection & fairness audits**:
    - Check outputs across sensitive groups (gender, race, age, socioeconomic status).
    - Use fairness metrics (demographic parity, equalised odds, disparate impact).
    - Propose mitigations (reweighting, thresholds, augmentation, post-processing).

Deliverable: "Bias & Fairness Analysis" section in the final report.

Step 6: Final Presentation & Communication

- **Two deliverables for mixed audiences:**

1. **Technical presentation** (Jupyter slides / LaTeX Beamer) → peers.
2. **Business-facing presentation** (PowerPoint / Canva) → executives (ROI, risks, strategy).
3. 8–12 slides per deck recommended.

Deliverables: Two slide decks (technical + business).

Step 7: GitHub Profile & Upload

- Create a public GitHub repo structured like an Open Source project:
- Include: src/ for scripts, notebooks/, data/, models/ directories.
- **Deliverables:** GitHub repo link + final report + reproducible code.

(Optional) Step 8: Deployment & MLOps
Complete this step if you have a good understanding of model deployment and MLOps practices.

- **Local deployment (required):** deploy the best model via **Flask**, **FastAPI**, or **Dash**.
- **Optional cloud:** AWS SageMaker, GCP Vertex AI, or Azure ML.
- **MLOps practices:**
    - Reproducible environments (requirements.txt, Docker).
    - Config-driven runs & experiment tracking (MLflow, W&B, etc.).
    - CI checks (lint, unit tests), basic monitoring plan.
    - Versioning & rollback plan.
- Provide a **demo** (GIF/screencast).

Deliverables: Running app + deployment guide + demo media.

(Optional) Step 9: Use of Generative AI

You may optionally use Generative AI tools to:

- Use LLMs to auto-generate EDA summaries or data dictionaries.
- Build GenAI-enhanced applications (e.g., LLM-backed recommenders, chatbots)

Deliverable:

- Document how Generative AI was used in the project
- Include code + examples in the GitHub repo or presentation
- Demo video.

Bonus points (up to +5) will be awarded for a creative and well-presented submission.

Pillar 5: Capstone Project: Rubric				
Criteria	**Ratings**			**Pts**
Step 1: Problem Understanding & Framing	**10 to >5.0 pts** **Outstanding/Exemplary** • Problem clearly framed with strong business context and data science perspective. • Task type (classification/regression/etc) correctly identified and justified. • Success metrics (technical + business KPIs) are relevant, measurable, and well-explained.	**5 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • Problem stated but lacks depth in business context or data science framing. • Task type identified but weakly justified. • Metrics mentioned but not well-aligned with problem or KPIs unclear.	**0 pts** **No Marks** • No clear problem statement or task type. • Metrics missing or irrelevant.	10 pts
Step 2: Data Collection & Understanding	**10 to >5.0 pts** **Outstanding/Exemplary** • High-quality dataset chosen and justified (source cited). • Comprehensive dataset overview: feature types, missing values, outliers, distributions. • Clear, complete data dictionary (variables, types, ranges/units).	**5 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • Dataset chosen, but justification is weak. • Overview provided but missing some key information (e.g., missing value summary or feature types). • Data dictionary is incomplete or lacks detail.	**0 pts** **No Marks** No explanation or incorrect distinctions.	10 pts

Step 3: Data Preprocessing, EDA & Feature Engineering	**10 to >5.0 pts** **Outstanding/Exemplary** • All preprocessing steps documented with reproducible code. • Clear handling of nulls, outliers, and duplicates. • Insightful applied EDA with visuals, distributions, and correlations. • Feature engineering shows domain knowledge and creativity. • At least one feature selection + dimensionality reduction method used and justified.	**5 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • Preprocessing done but not fully explained. • Basic EDA with few insights or missing relationships. • Feature engineering minimal (basic encoding/scaling only). • Feature selection or dimensionality reduction was used but not justified.	**0 pts** **No Marks** • No meaningful preprocessing or EDA done. • No feature engineering/selection applied.	10 pts
Step 4: Model Implementation & Comparison	**20 to >10.0 pts** **Outstanding/Exemplary** • Multiple models implemented and tuned appropriately. • Evaluation metrics correctly applied and compared across models. • Reproducibility ensured (saved models/configs). • Clear reasoning for model choice based on results.	**10 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • One or two models implemented with minimal tuning. • Evaluation metrics applied but comparison weak or incomplete. • Limited explanation of model choice.	**0 pts** **No Marks** • No working model implementation. • No evaluation or comparison provided.	20 pts
Step 5: Critical Thinking, Ethical AI & Bias Auditing	**20 to >10.0 pts** **Outstanding/Exemplary** • Excellent use of explainability tools (SHAP/LIME/PDP/ICE). • Thorough discussion of data/model limitations (imbalance, leakage, overfitting). • Bias audit performed across sensitive groups with fairness metrics. • Proposes clear, feasible mitigation strategies.	**10 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • Some attempt at explainability but limited insights. • Mentions model limitations superficially. • Bias or fairness checks present but incomplete (no metrics or no mitigations).	**0 pts** **No Marks** • No bias audit, fairness consideration, or explainability.	20 pts

Step 6: Final Presentation & Communication	**10 to >5.0 pts** **Outstanding/Exemplary** • Two high-quality, well-structured presentations (technical + business). • Technical deck: clear methodology, visuals, metrics. • Business deck: ROI, risks, strategy clearly communicated for a non-technical audience. • Visually professional and concise (8–12 slides per deck).	**5 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • Presentations provided but lack polish or depth. • Either technical or business deck missing key elements. • Weak storytelling or cluttered slides.	**0 pts** **No Marks** • No presentations delivered. OR • Slides fail to communicate key findings.	10 pts
Step 7: GitHub Profile & Upload	**15 to >10.0 pts** **Outstanding/Exemplary** • Public GitHub repo structured like an open-source project (src/, notebooks/, data/, models/). • Includes README, requirements.txt, final report, reproducible code. • Clean, professional commit history.	**10 to >5.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** • GitHub repo provided but lacks structure or documentation. • Code not fully reproducible or poorly organised.	**5 to >0 pts** **No Marks** • No GitHub repo shared. • Code and report missing.	15 pts
Bonus points Creative and well-presented submission	**5 to >2.5 pts** **Outstanding/Exemplary** Demonstrates exceptional creativity, originality, and clear presentation. The submission goes beyond expectations in terms of design, clarity, or innovation.	**2.5 to >0.0 pts** **Adequate / Meets Basic Expectations (Partial Points)** Shows some creativity and effort in presentation but lacks depth or refinement in execution.	**0 pts** **No Marks** Minimal creativity or unclear presentation. No additional effort demonstrated beyond the basic requirements.	5 pts
				Total Points: 100